Here, $15 = \frac{360}{r!}$

i.e., $r! = \frac{360}{15} = 24$

$r! = 4 \times 3 \times 2 \times 1$

$r! = 4!$

$\therefore \quad r = 4$

Example 7.14

If $nC_8 = n\,C_{7,}$ find $n\,C_{15}$

Solution:

$$nC_8 = n\,C_7$$
$$n\,C_{n-8} = nC_7$$
$$n-8 = 7$$
$$n = 15$$

Now, $nC_{15} = 15C_{15} = 1.$

7.4 Introduction to Binomial, Exponential and Logarithmic series

7.4.1 Binomial series

A binomial is an algebraic expression of two terms. Now let us see the following binomial expansion and the number pattern we get adjacent to it.

Binomials	Expansions	Number pattern
$(x+a)^0$	1	1
$(x+a)^1$	$x+a$	1 1
$(x+a)^2$	$x^2+2xa+a^2$	1 2 1
$(x+a)^3$	$x^3+3x^2a+3xa^2+a^3$	1 3 3 1
$(x+a)^4$	$x^4+4x^3a+6x^2a^2+4xa^3+a^4$	1 4 6 4 1
$(x+a)^5$	$x^5+5x^4a+10x^3a^2+10x^2a^3+5xa^4+a^5$	1 5 10 10 5 1

From the above, we observe that the binomial coefficients form a number pattern which is in a triangular form. This pattern is known as Pascal's triangle. [In pascal's triangle, the binomial coefficients appear as each entry is the sum of the two above it.]